

Smart Attendance System Using BLE Beacons & Flutter App

Milestone 1: Setup Basic Flutter App and BLE Beacons

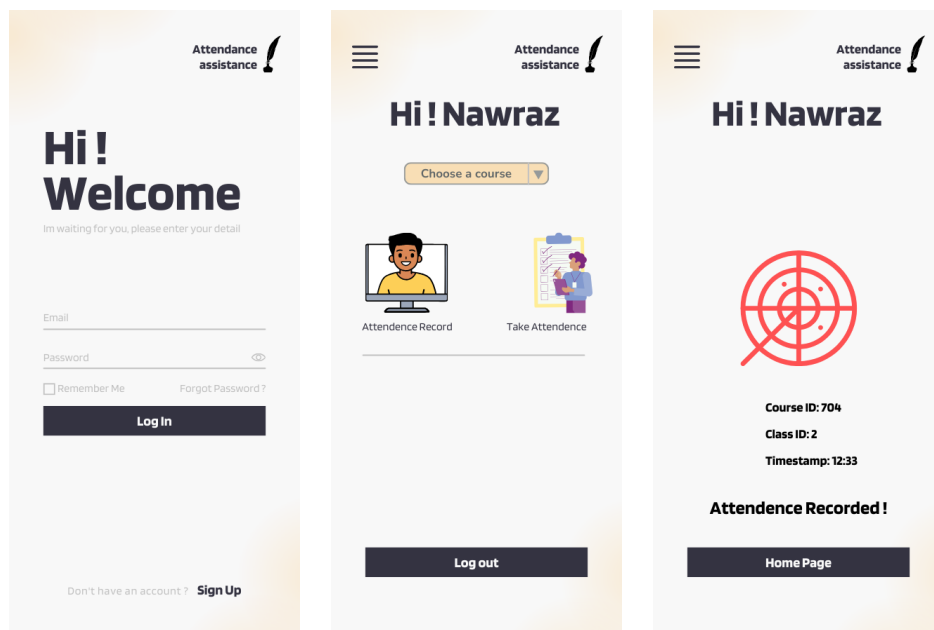
Objective: Create a basic Flutter app to detect beacon signals.

Requirements:

1. Build a Basic Flutter App :

- Set up a Flutter project with student authentication.
- Implement a simple login (e.g., using Firebase Authentication) to verify students.
- Integrate a BLE scanning package (e.g., flutter_blue_plus).
- Implement functionality to scan for nearby BLE beacons.
- Display the detected beacon's UUID and signal strength (RSSI) on the app screen.
- Enable each student to monitor their attendance level at each class.

Main home page should be some thing like this :



2. Deploy BLE Beacons :

- Assign a unique UUID to each beacon for identification.
- Adjust the signal strength (RSSI) to define the detection range (e.g., 5-10 meters)

Milestone 2: Implement Attendance Logging and Cloud Integration

Objective: Enable the app to log attendance and send data to the cloud.

Requirements:

1. Log Attendance :

- When a beacon is detected, log the student's ID, class ID, beacon UUID, and timestamp.
- Store this data to firebase database and ensure the app can sync data when online.

2. **Basic Dashboard :**

- Set up a simple dashboard for each student to view his/her attendance logs of each course.
- Show attendance percentages and define the absence level.

Important Notes:

- You will be provided a more detailed description for each milestone, specifying submissions guidelines and deadline.
- This project holds 25% of the course, evenly distributed across the two milestones.
- You can work in teams of 3 minimum and 5 maximum. Kindly submit your team info through this [link](#).